Replay16 T1: LibriSpeech gains, mixed domain results

Same-step comparison at optimizer step 600; lower validation p_err is better.

Dataset	No replay	BCR16 T1	Delta (pp)
OpenML bad	7.9695%	7.9354%	-0.0341
Earnings (6 parents)	4.4560%	4.4755%	+0.0195
LibriSpeech clean	1.2056%	1.1603%	-0.0453
LibriSpeech other	2.6476%	2.5681%	-0.0795

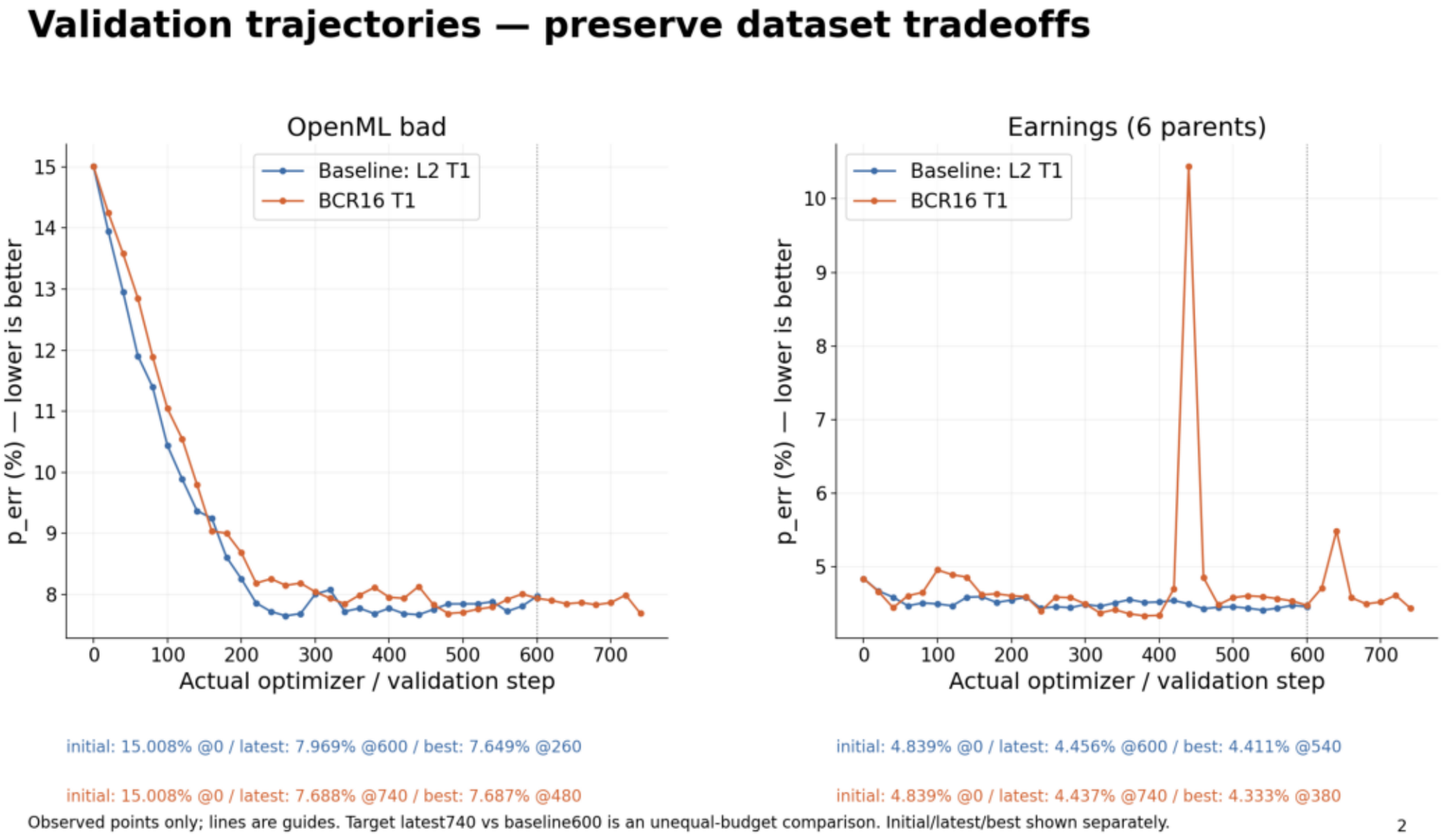
What changed

128 fresh prompts → 112 fresh + up to 16 replay; buffer 400, seed 42.
Reward divisor also changes 4 → 3: this is not an isolated replay ablation.
Shared: ReMax, temperature 1, LoRA 320, LR 3e-6 constant, 8 rollouts, 16 GPUs.
Recoveries reset replay; equal steps do not imply equal fresh-data exposure.

Baseline: d807d91db6c775c7d097c5ef7ae2f734 | code 03680b38dd
Target: 73acea8095b222e76c04affb7133c92e | code 1d72d5a4a5 | Both stopped

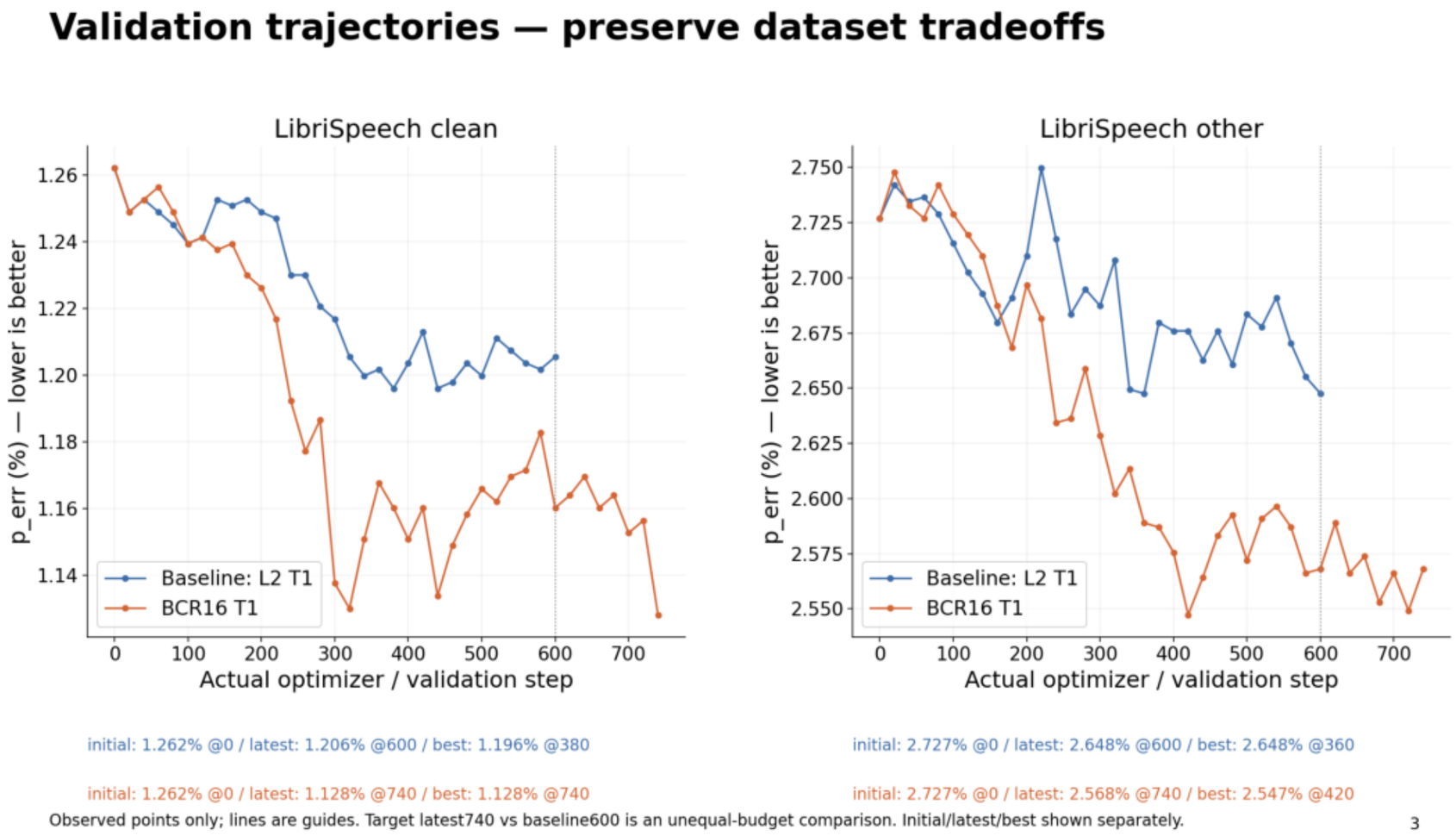
Validation trajectories: keep dataset tradeoffs visible

Observed validation points; latest target step 740 versus baseline 600 has unequal budgets.



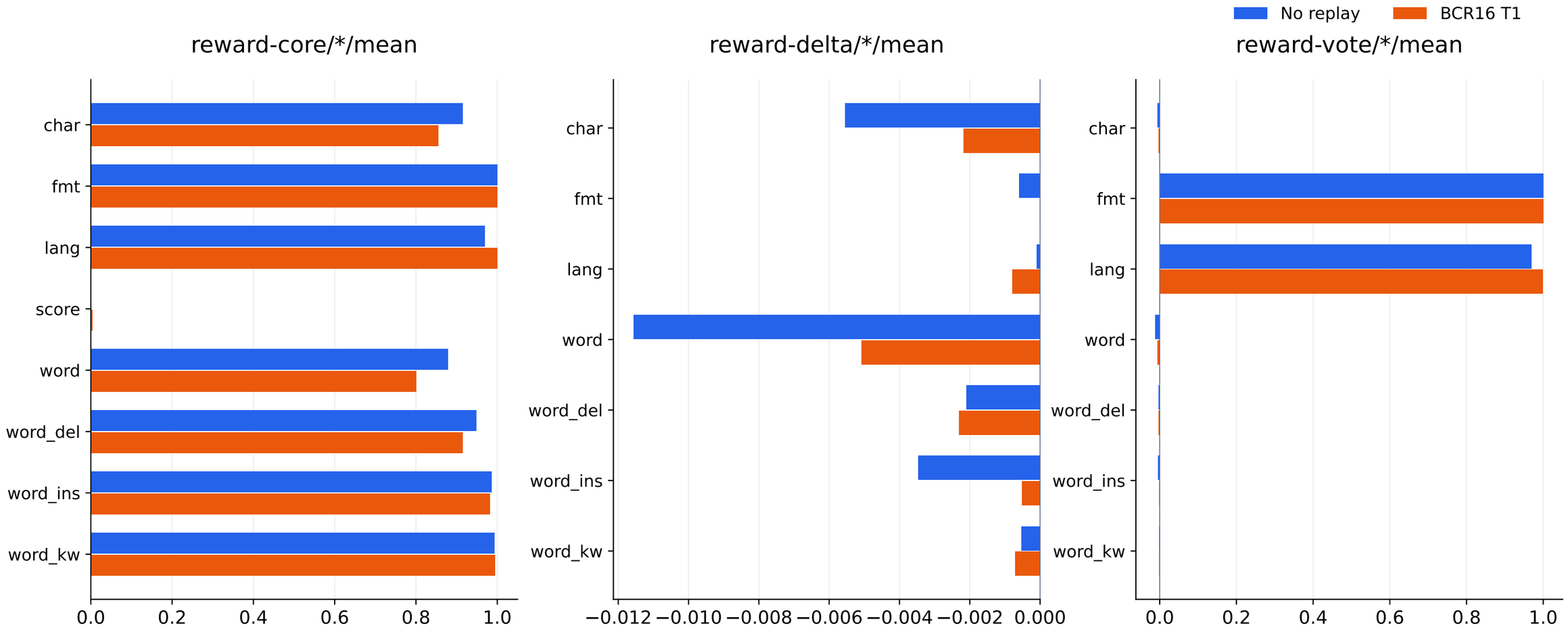
Validation trajectories: keep dataset tradeoffs visible

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Training reward means: diagnostics, not quality rankings

Exact shared optimizer steps 591–600; arithmetic mean, 10 observations per run and component.



Core score uses different divisors (4 vs 3). Batch composition differs; reward increases do not establish better ASR.
This compact version omits raw reward traces; they remain in the full PDF.

Matched examples: audit normalization before tuning rewards

Step 600; stable identities and identical references. Official scorer replay reproduced 12/12 selected cases.

Dataset	Matched units	Improved / worsened	Net added errors
OpenML bad	393	27 / 20	-2
Earnings	6 parents (341 chunks)	2 / 4	+3
LS clean	2,620	50 / 28	-24
LS other	2,939	99 / 62	-42

Three illustrative scoring-sensitive changes

OpenML 11746087412974645092: “Engel, 2006” → “Engel (2006)”; errors 2 → 6.
LS other 6070-86745-0002: “9:45” → “a quarter to ten”; errors 4 → 0.
LS clean 3729-6852-0046: “altogether” → “all together”; errors 0 → 2.

Next check: audit hypothesis-dependent normalization on the same matched corpus.
Two OpenML samples change n_ref. Cases do not establish prevalence or acoustic causes.
Keep official per-dataset p_err and Earnings guardrails; reward and exposure changes limit causal claims.